

Supplementary Information for “LLM-Empowered Decision-focused Learning for the Operation of Local Energy Communities”

Yangze Zhou¹, Yu Zuo¹, Daniel Kirschen², Yi Wang^{1*}

¹Department of Electrical and Electronic Engineering, The University of
Hong Kong, Hong Kong SAR, China.

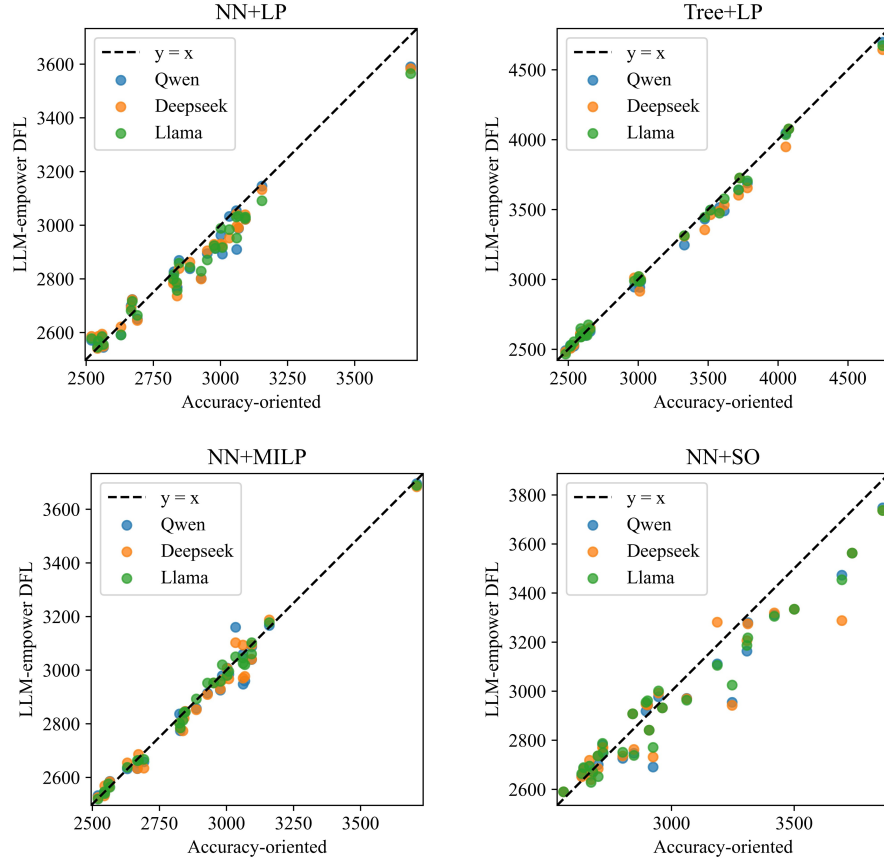
²Department of Electrical and Computer Engineering, University of
Washington, Seattle , United States.

*Corresponding author(s). E-mail(s): yiwang@eee.hku.hk;
Contributing authors: yzzhou@connect.hku.hk;
yzuo1998@connect.hku.hk; kirschen@uw.edu;

Supplementary experimental result

0.1 Statistical Visualization of the Cost Reduction

The cost comparison between the LLM-empowered DFL and the accuracy-oriented method for each test sample is shown in Fig. 1. The x and y -axes represent the costs achieved by the accuracy-oriented baseline and the LLM-empowered DFL, respectively. Across all four settings, most points fall below the $y = x$ reference line, indicating that the LLM-empowered approach generally attains lower objective values than the accuracy-oriented baseline on the majority of instances. This distribution suggests consistent performance gains rather than random fluctuations around parity.



Supplementary Fig. 1: The cost comparison of the LLM-empowered DFL with the accuracy-oriented method.

0.2 Comparison to Heuristic Method

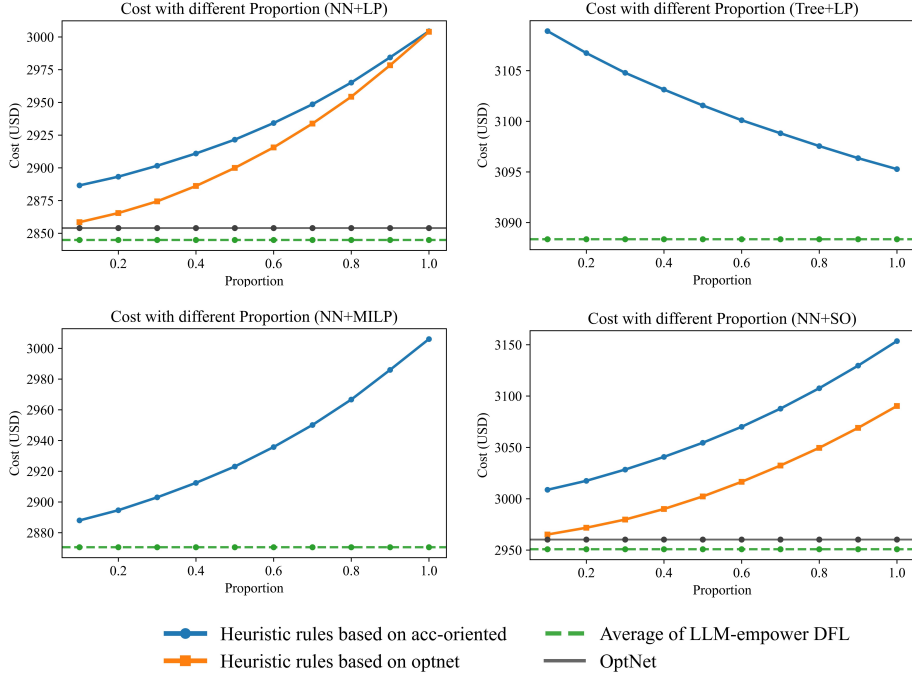
This section provides a comparison with heuristic rule-based methods. Given the selected similar historical samples in the training set, denoted as $\hat{y}^{\text{train}}$ and y^{train} , we first compute the historical forecast errors

$$e = y^{\text{train}} - \hat{y}^{\text{train}} \quad (1)$$

We then heuristically correct the current-day forecast $\hat{y}$ by adding a proportion of this historical error:

$$\tilde{y} = \hat{y} + \alpha e \quad (2)$$

where α is a tunable proportion coefficient (grid-searched in our experiments). The corrected forecast $\tilde{y}$ is subsequently fed into the same downstream optimization module, and the resulting operational cost is reported for comparison.



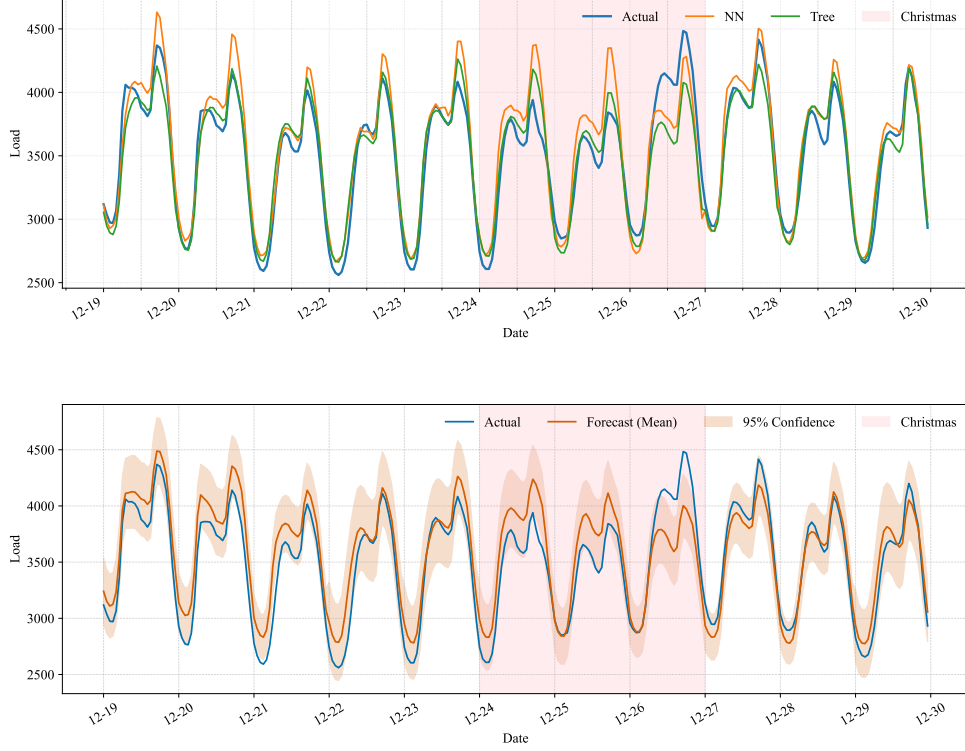
Supplementary Fig. 2: The cost comparison of the LLM-empowered DFL with the heuristic rules-based method.

The cost comparison of the LLM-empowered DFL with the heuristic rules-based methods has been given in Fig. 2. We evaluated the effectiveness of applying these heuristic rules to both original forecasts. Our findings indicate that when an NN serves as the forecasting model, directly using these errors to fine-tune forecasts via

heuristic rules unexpectedly led to higher costs. This suggests that enabling the forecasting model to understand downstream decision-making information for fine-tuning purposes is a more effective approach. While the heuristic rules-based method proved effective when tree-based models were used for forecasting, their performance did not surpass that of our LLM-empowered DFL framework.

0.3 Performance under Out-of-Distribution

Fig. 3 illustrates the forecasting dynamics during the Christmas period. Influenced by the holiday effect, the actual load plummeted on Dec. 24 and Dec. 25. Traditional models failed to capture this abrupt shift, significantly overestimating the load during the drop. Furthermore, as industrial activities resumed on Dec. 26, these models exhibited a severe lag, underestimating the rebound because their input features were dominated by the suppressed load values from the preceding holiday days.



Supplementary Fig. 3: (a) The deterministic forecasting results from 19 Dec. to 29 Dec 2013. (b) Actual load and Probabilistic Forecasts with 95% intervals from 19 Dec. to 29 Dec 2013.

As quantified in Table 1, our LLM-empowered DFL approach achieves lower operational costs compared to Accuracy-oriented FL and OptNet. Notably, these results were obtained without explicit “holiday” labeling in the input prompts or specialized preprocessing for similar-day searching and few-shot learning. This confirms that our proposed method exhibits robust zero-shot adaptability even in the absence of such guidance.

Supplementary Table 1: The average daily costs for 4 settings during Christmas (USD)

	NN+QP/LP	Tree+QP/LP	NN+MIQP/LP	NN+SO
Ideal	2452.93	2452.93	2455.64	2452.931
Acc-oriented	3126.45	3105.89	3126.93	3141.49
OptNet	3091.38	/	/	3139.57
Qwen	3088.29	3102.76	3126.12	3134.07
Deepseek	3068.08	3106.38	3111.05	3138.00
Llama	3073.85	3098.85	3111.41	3131.02

Beyond quantitative metrics, a key advantage of our framework is its flexibility via prompt engineering. Unlike rigid numerical models, it facilitates the explicit injection of domain knowledge. For instance, observed error patterns, such as the specific overestimation and underestimation trends during holidays, can be described in the prompt as prior experience. This enables the LLM to adjust its strategy to compensate for deviations. Similarly, for other OOD scenarios such as rapid load spikes caused by extreme weather, relevant contextual experiences can be fed into the LLM, thereby significantly enhancing the method’s robustness where traditional models might fail.